COAL Assignment #02

Date: / /

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Section:- BAI-3A

Q1 Multiply Ax by 26 Ax = 3

$26 = 16 + 8 + 2$

$(3 \times 26) = (3 \times 16) + (3 \times 8) + (3 \times 2)$

```
mov Ax, 3
mov Bx, Ax
shl Bx, 4

mov Cx, Ax
shl Cx, 3

shl Ax, 1
add Ax, Bx
add Ax, Cx
```

Q2

```
.data
num DWORD 2345ABCDh
.code
    mov esi, OFFSET num
    mov ecx, 4
  Right shift:
      ror BYTE PTR [esi], 1
      inc esi
      Loop Right-Shift:
```

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```
mov esi OFFSET num
mov ecx,4
LeftShift:
    rol BYTE PTR [esi],1
    inc esi
    loop LeftShift
```

Q3

```
.data
    num DWORD 2345ABCDh
.code
    mov esi, OFFSET num
    mov ecx, 2
RightShift:
    ror WORD PTR [esi], 1
    add esi,2
    loop RightShift
mov esi, OFFSET num
mov ecx,2
LeftShift:
    rol WORD PTR [esi], 1
    add esi, 2
    loop LeftShift
```

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Q4

a)
```
.data
Bseconds BYTE ?
.code
mov dx, 0101100111 011010b
mov ax, dx
and ax, 001Fh
mov Bseconds, al
```

b)
```
.data
Bminutes BYTE ?
.code
mov dx, 0101100111011010b
mov ax, dx
shr ax, 5
mov Bminutes, al
```

Q5

	Instruction	ax	dx	CF
1	mov dx, 9A3Fh		9A3Ff	
	mov ax, 7580h	75		
1	mov dx, 9A3Fh	ax = - -	dx = 9A3Fh	CF = -
2	mov ax, 7580h	ax = 7580h	dx = 9A3Fh	CF = -
3	clc	ax = 7580h	dx = 9A3Fh	CF = 0
4	add ax, 90EAh	ax = 106Ah	dx = 9A3Fh	CF = 1
5	adc dx, 1	~~ax = 106~~	ax = 106Ah	dx = 9A41h CF = 0
6	stc	ax = 106Ah,	dx = 9A41h	CF = 1
7	add ax, 8120h	ax = 918Ah	dx = 9A41h	CF = 0
8	adc dx, 0FFFFh	~~ax = 918Ah~~		
	ax = 918Ah	dx = 9A40h		CF = 1

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Q6

a) Test ax, ax
b) Test bx, 1
c) Test dx, dx
d) Test dx, 8000h
e) test bl, bl

Q7

```
mov dx, 5
stc          CF = 1
mov ax, 10h  ax = 16 (in decimal)
adc dx, ax   dx = 5+16+1 = 22
```

Q8

1 Use OFFSET to load val1 and val2 addresses not their values.
```
mov esi, OFFSET val1
mov edi, OFFSET val2
```
2 mov BYTE PTR [esi], al overwrites val1 store it through result Pointer (use EBX)
3 replace DEC with INC to move forward because data is stored in little endian format.
4 add inc ebx to move result Pointer

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Q9

```
INCLUDE Irvine32.inc
.data
  msg BYTE "Enter marks (0-100): ",0
  invalidMsg BYTE "Invalid Marks",0
  grade BYTE "Grade: ",0

.code
main PROC
    mov edx, OFFSET msg
    call WriteString
    call ReadInt
    cmp eax, 0
    jl InvalidMarks
    cmp eax, 100
    jg InvalidMarks
    mov edx, OFFSET grade
    call WriteString
    cmp eax, 80
    jae GradeA
    cmp eax, 60
    jae GradeB
    cmp eax, 40
    jae GradeC
    jmp GradeF
GradeA:
    mov al, 'A'
    jmp Display
GradeB:
    mov al, 'B"
    jmp Display
```

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```
GradeC:
    mov al, 'C'
    jmp Display
GradeF:
    mov al, 'F'
    jmp Display
Display:
    call WriteChar
    call crlf
    jmp finish

InvalidMarks:
    mov edx, OFFSET InvalidMsg
    call WriteString
    call crlf
Finish:
    exit
main ENDP
END MAIN
```

Q10

Instruction	EAX	EBX	Stack
mov eax,3	3	-	Empty
mov ebx,7	3	7	Empty
push eax	3	7	3
push ebx	3	7	7,3
pop eax	7	7	3
push 10	7	7	10,3
pop ebx	7	10	3
pop eax	3	10	Empty

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Q11

```
.data
num1 SDWORD ?
num2 SDWORD ?
num3 SDWORD ?
num4 SDWORD ?
result SDWORD ?
.code
main PROC
mov eax, num1
cmp eax, num2
jle else_if

mov ebx, num3
cmp ebx, num4
je elseif

mov eax, num1
add eax, num3
mov result, eax
jmp endif

mov eax, num2
cmp eax, num4
jne else

mov eax, num2
sub eax, num1
mov result, eax
jmp endif
```

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elseif:
mov eax, num2
cmp eax, num4
jne else

mov eax, num2
sub eax, num1
mov result, eax
~~jump~~ jmp endif

else:
move result, 0

endif:
~~ret~~ exit
main ENDP
END main

Q12 .data
op1 SDWORD ?
op2 SDWORD ?
x SDWORD ?
y SDWORD ?
z SDWORD ?

.code
main PROC
while Loop:
mov eax, op1
cmp eax, op2
jg Finish

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```
mov eax, op1
cmp eax, x
jl else

comp eax, y
jg else

add z, 10
jmp next

else:
  sub z, 10

next:
  dec op1
  jmp whileLoop

Finish:
  ret

main ENDP
END main
```

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